

LAB 1

Experimental Protocols

Testing the Fairness of a Dice

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Testing the Fairness of a Sanded Dice

Protocol: Sanded the face opposite the six (the '1') with sandpaper, then rolled the altered die 40 times.

Comparison: Simulated 40 rolls of an unaltered die in R as a benchmark for a normal die.

Hypotheses ($\alpha = 0.05$, two-tailed): $H_0: P(6) = 1/6$ and $E(X) = 3.5$ (die is fair), $H_a: P(6) \neq 1/6$ and/or $E(X) \neq 3.5$



Justification for the Design

Why we altered the die, ran 40 trials, and chose these tests

1

Sanding the face opposite the six

The '1' face contacts the table when a six lands face-up. Adding friction there can tip borderline rolls toward six.

2

40 trials

A subtle alteration needs more rolls to reveal any real bias, and more trials pull the sample mean closer to the true mean (Law of Large Numbers), averaging out uncontrolled factors like roll technique.

3

Two-tailed test at $\alpha = 0.05$

We test for unfairness in either direction (not just 'low probability of 6'), using probability and mean together. $\alpha = 0.05$ is conventional and balances Type I vs. Type II error.

Results & Analysis

40 rolls of the altered dice, tested against a fair die's expected values

TESTING THE MEAN (ONE-SAMPLE T-TEST)

3.25

sample mean · expected $\mu_0 = 3.5$

t-statistic: -0.952

p-value: 0.347

95% CI: (2.72, 3.78) — contains 3.5

TESTING P(SIX) (BINOMIAL TEST)

12.5%

observed rate · expected $P(6) = 16.7\%$

Frequency of 6: 5 of 40 rolls

p-value: 0.670 (exact, two-sided)

95% CI: (4.2%, 26.8%) — contains 1/6

Both p-values exceed $\alpha = 0.05$, and both confidence intervals contain the fair-dice benchmark.

Conclusion & Confidence

Fail to reject H_0 — no evidence (yet) the die is unfair

Mean test

$$p \approx 0.347 > 0.05$$

Fail to reject $E(X) = 3.5$

P(6) test

$$p \approx 0.670 > 0.05$$

Fail to reject $P(6) = 1/6$

Joint result

Both tests fail to reject

Dice 1 is not shown to be unfair

Confidence in the result:

- The 95% CI for the mean, (2.72, 3.78), and for P(6), (4.2%–26.8%), both comfortably contain the fair-dice values, consistent with the failed rejections.
- Limitation: with only 40 trials, this test has limited power, a small, real bias could still go undetected at $\alpha = 0.05$.

Next Steps: Modifications for EP2

EP1 failed to reject fairness, EP2 uses a more drastic, targeted alteration (adding weight to the '1' face)

CHANGING

- **Alteration method:** adding weight (double-sided tape) to the '1' face instead of sanding, a more drastic physical bias.
- **Hypothesis form:** switch to a one-tailed test, since EP2 predicts a specific direction, more sixes, not just 'different' than a fair dice.
- **Alternative hypotheses:** $P(6) > 1/6$ and $E(X) > 3.5$.

STAYING THE SAME

- Focus on a single face (6) for precise, intentional comparisons.
- Same fair-die benchmarks: $P(6) = 1/6$ and $E(X) = 3.5$.
- Same sample size: 40 trials, for direct comparability with EP1.
- Same significance level: $\alpha = 0.05$, the conventional standard.



Thank You

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